Book strip

Science · GROW · Autumn 2 · Week 1 · GROW_SCI_A2_W01_P2

Day and night: show what you know · Weekly arc: Show what you know.

Goal. I can use a rotating-Earth model to explain day, night, and apparent Sun motion.

Today I can. I can order four positions of one place across a 24-hour rotation cycle. I can link each position to light, shadow, and the Sun's apparent movement. I can use model observations to refute a misconception and state one model limit.

Made by Matt

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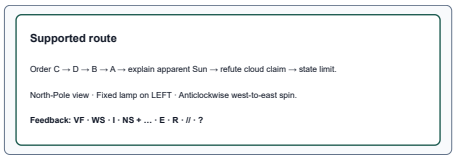
Made by Matt

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Route card

Staff: cut these cards apart before the lesson. Give one route card. Mark actual **I · S · C** beside each task.

Supported



Supported route card. Order the four rotation frames, explain apparent Sun motion, and state a limit.

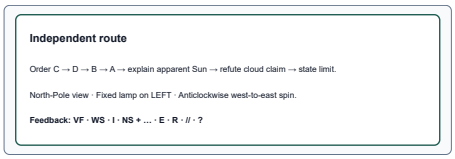
Use Reference A. Order Frames C, D, B, and A starting at sunrise (Top). Match each frame to entering light, facing light, leaving light, or facing away.

Use Reference A and Reference B. Work in your book or usual mode.

Staff I ☐ S ☐ C ☐ · Feedback: VF ☐ WS ☐ E ☐ NS + … ☐ R ☐ // ☐ ? ☐

My response:

Independent



Independent route card. Order the four rotation frames, explain apparent Sun motion, and state a limit.

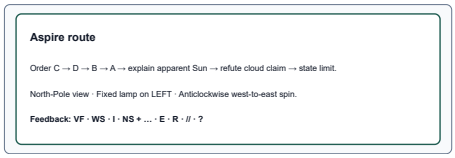
Use Reference A. Complete a four-row evidence record for marker X (Frames C, D, B, and A). State the position, illumination state, and apparent Sun position.

Use Reference A and Reference B. Work in your book or usual mode.

Staff I ☐ S ☐ C ☐ · Feedback: VF ☐ WS ☐ E ☐ NS + … ☐ R ☐ // ☐ ? ☐

My response:

Aspire



Aspire route card. Order the four rotation frames, explain apparent Sun motion, and state a limit.

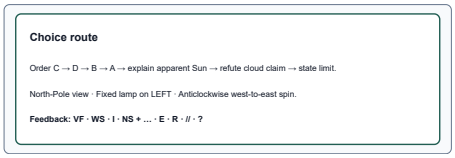
Use Reference A. Construct a four-frame North-Pole evidence account for marker X. Link anticlockwise rotation to apparent east-to-west solar motion.

Use Reference A and Reference B. Work in your book or usual mode.

Staff I ☐ S ☐ C ☐ · Feedback: VF ☐ WS ☐ E ☐ NS + … ☐ R ☐ // ☐ ? ☐

My response:

Choice



Choice route card. Order the four rotation frames, explain apparent Sun motion, and state a limit.

1. Look at Reference A. Which frame shows midday facing the lamp: Frame D (Left) or Frame A (Right)? 2. Which frame shows night facing away: Frame A or Frame D?

Use Reference A and Reference B. Work in your book or usual mode.

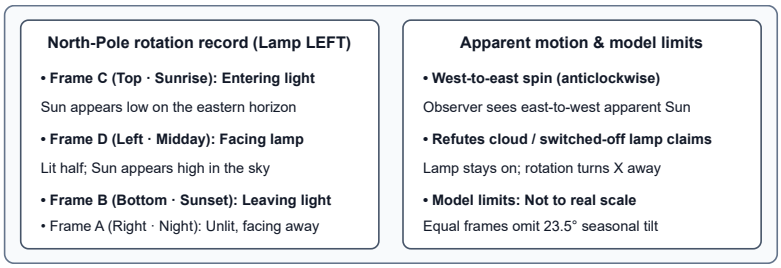
Staff I ☐ S ☐ C ☐ · Feedback: VF ☐ WS ☐ E ☐ NS + … ☐ R ☐ // ☐ ? ☐

My response:

Made by Matt

Reference sheets

Reference A · Four-frame North-Pole rotation record (Fixed lamp on LEFT)

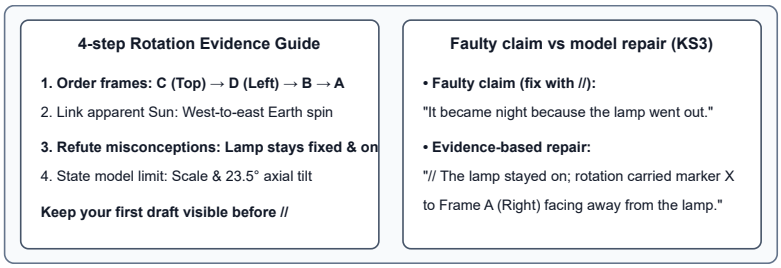


North-Pole rotation record. It links Frames C, D, B, and A to apparent Sun motion and limits.

Reference only. Do not write on this sheet. Use these supplied observations to support your rotation explanation.

Ordered frame & position.	Illumination state of marker X.	Apparent Sun position from marker X.	Rotation clue & model note.
1. Frame C · Top (Sunrise).	Entering the lit left half from shadow.	Sun appears low on the eastern horizon.	Start of cycle (0°); Earth spins anticlockwise (west to east).
2. Frame D · Left (Middle of day).	Facing the fixed lamp directly in the lit half.	Sun appears high across the sky.	After 1st quarter turn (90°); lamp stays fixed and on.
3. Frame B · Bottom (Sunset).	Leaving the lit left half into shadow.	Sun appears low on the western horizon.	After 2nd quarter turn (180°); observer turns away from lamp.
4. Frame A · Right (Night).	Facing away from the lamp in the unlit right half.	Sun is below the local horizon (darkness).	After 3rd quarter turn (270°); 4th turn returns X to Top (360°).

Reference B · Refuting misconceptions, apparent motion, and model limits



Four-step Rotation Evidence guide. It shows frame ordering, apparent motion, refuting cloud claims, and limits.

Evidence step.	Sentence stem for your book.	Key Stage 3 check.
1. Order the 24-hour cycle.	From sunrise, marker X moves through C (Top), D (Left), B (Bottom), and A (Right).	Did you track the same marker back to its starting point?
2. Explain apparent Sun motion.	Earth rotates west to east, so the Sun appears to move east to west.	Did you distinguish observer viewpoint from real solar motion?
3. Refute 'clouds/lamp off' claims.	No clouds are present and the lamp stays on. Rotation turns X	Did you use Frame D and Frame A to prove orientation changed?

Evidence step.	Sentence stem for your book.	Key Stage 3 check.
	into shadow.	
4. State one honest model limit.	// Limit: size and distance are not to scale. Equal frames omit 23.5° tilt.	Did you state what equal quarter turns cannot show?

Made by Matt

Catch-up sheet

Absence or access only. This sheet gives a fast entry if a pupil joined late. Never add it on top of today's routine tasks.

1. **Four rotation frames (Reference A).** View Earth above the North Pole with the lamp on the left. Track marker X anticlockwise: **C (Top)**, **D (Left)**, **B (Bottom)**, **A (Right)**.
2. **Why night happens.** The lamp stays on and there are no clouds. Earth's rotation carries marker X to the unlit right side.
3. **Model limit.** Classroom size and distance are not to scale. Equal frames omit seasonal tilt.
4. **Quick link.** Point to Reference A and B. Then join today's You do Task 1.

Point, speak, sign, draw, type, write, or tell an adult. You may pass.

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Choice route sheet

Circle or point to your choice. Give a short reason from Reference A and Reference B.

1. 1. Look at Reference A. Which frame shows midday facing the lamp: Frame D (Left) or Frame A (Right)? 2. Which frame shows night facing away: Frame A or Frame D? **Staff I ☐ S ☐ C ☐**

Help: Check Rows 2 and 4 on Reference A.

2. 3. What moves in this model: rotating Earth or the fixed lamp? 4. Why does the Sun seem to cross our sky: we rotate or Sun stops? **Staff I ☐ S ☐ C ☐**

Help: Check Row 3 on Reference A.

3. 5. Look at Reference B. Does the model need clouds to make night: No or Yes? 6. Name one model limit: not to scale or real space size? Add one // edit. **Staff I ☐ S ☐ C ☐**

Help: Check Reference B.

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Choice

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt